

- Q.28 Write difference between T and π Network.
 Q.29 Write limitations of prototype filters.
 Q.30 Write general characteristics of attenuators.
 Q.31 Write short note on symmetric network.
 Q.32 Write properties of piezoelectric filters.
 Q.33 What is SWR and VSWR.
 Q.34 What is the concept of High pass filters.
 Q.35 Write characteristics of LPF.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Write basic concept of active filters and their comparison with passive filters.
 Q.37 Write concept and advantage, disadvantage and area of application of HVDC.
 Q.38 Write a short note on following:
 a) Stubs b) Ladder Network

No. of Printed Pages : 4
 Roll No.

181035/171041/
 121041/031041

3rd Sem / Electronics and Power Eltx Sub : Network Filter and Transmission Lines

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The value of one decibel is equal to?
 a) 0.115 N b) 0.125 N
 c) 0.135 N d) 0.145 N
- Q.2 An RC coupling circuit is an example of what type of filter?
 a) Low pass filter b) High pass filter
 c) Band pass filter d) All pass filter
- Q.3 The expression of the open circuit impedance Z_{oc} is?
 a) $Z_{oc} = Z_1/2 + Z_2$ b) $Z_{oc} = Z_2/2 + Z_2$
 c) $Z_{oc} = Z_1/2 + Z_1$ d) $Z_{oc} = Z_1/2 - Z_2$
- Q.4 The leakage current in the transmission lines is referred to as the
 a) Resistance b) Radiation
 c) Conductance d) Polarization
- Q.5 The cut off frequency of the constant k-low pass filter is?
 a) $1/\sqrt{LC}$ b) $1/(\pi\sqrt{LC})$
 c) $\sqrt{LC}$ d) $\pi\sqrt{LC}$

Q.6 If V_1 is the voltage at port 1 and V_2 is the voltage at port 2, then the attenuation in dB is?

- a) $20 \log_{10}(V_1/V_2)$ b) $10 \log_{10}(V_1/V_2)$
c) $20 \log_{10}(V_2/V_1)$ d) $10 \log_{10}(V_2/V_1)$

Q.7 The expression of the characteristic impedance of a symmetrical T-Section is?

- a) $Z_{OT} = \sqrt{(Z_1^2/4 - Z_1 Z_2)}$
b) $Z_{OT} = \sqrt{(Z_1^2/4 - Z_1)}$
c) $Z_{OT} = \sqrt{(Z_1^2/4 - Z_2)}$
d) $Z_{OT} = \sqrt{(Z_1^2/4 - Z_1 Z_2)}$

Q.8 Which of the following parameters is not a primary parameter?

- a) Resistance b) Attenuation constant
c) Capacitance d) Conductance

Q.9 What is a filter?

- a) Frequency selective circuit
b) Amplitude selective circuit
c) Frequency damping circuit
d) Amplitude damping circuit

Q.10 The line having R, L, C distributed along the circuit are called

- a) Lumped b) Distributed
c) Parallel d) Paired

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

Q.11 Network is a combination of _____ element.

Q.12 Write full form of LPF

Q.13 Decibel is the unit of _____.

Q.14 Draw Pi type network.

Q.15 Write full form of HVDC.

Q.16 VSWR means _____.

Q.17 What is network.

Q.18 Write any two names of active component.

Q.19 Resistor is a _____ components.

Q.20 What is balance type network.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

Q.21 Differentiate between unilateral and bilateral network.

Q.22 Give five applications of transmission line.

Q.23 List five applications of attenuators.

Q.24 Explain the construction and working of double crystal BPF.

Q.25 Explain briefly insertion loss.

Q.26 Write short note on iterative impedance.

Q.27 What is need of m-derived filters?